Part –IV

Answer in detail **(10 Marks)**

1. State the main divisions and general classification of surveying.
2. What do you mean by ranging? Explain the types of ranging.
3. Explain with a neat diagram the construction and working of an optical square.
4. Explain the temporary adjustments of a dumpy level.
5. The level readings taken are listed as follows:

   1.430, 2.015, 1.005, 0.400, 3.370, 2.975, 1.415, 0.695

   The instrument is shifted after the 4th and 8th readings. The RL of Ist point is 100.000 m. Enter the reading in a field book form and reduce the RL by HC Method. Apply usual check.
6. The level readings taken are listed below:

   4.390, 7.620, 6.520, 3.910, 5.390, 4.730, 6.290, 3.520, 4.330, 2.990.

   The RL of First point is 200.000 m. The instrument is shifted after 3rd and 8th points. Enter the reading in a field book form and reduce RL by Rise and fall method. Apply usual check.

Answers:

1) b 2) c 3) b 4) b 5) a 6) a 7) a

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